PROJECT ATO / DESIGN DOSSIER 02

A quantitative architecture for invasive spinal interfaces

Selected resources, timing, output limits and physical allocations

8 October 2026 · Entirely digital · Proposed system under development

ATO links spinal recording, causal interpretation, learner-specific stimulation and feedback. This dossier gives the proposed implant a finite resource architecture. It preserves both expert-informed motor practice and selected physiological automation.

Reference configuration	Causal purpose
Paced multidirectional foot placement	Expert spinal representation → learner calibration → active practice with timed assistance → unassisted retention assessment
External systolic-pressure feedback after SCI	Qualified epidural pressor profile → measured SBP → bounded feedback; optional spinal biomarker must show added value

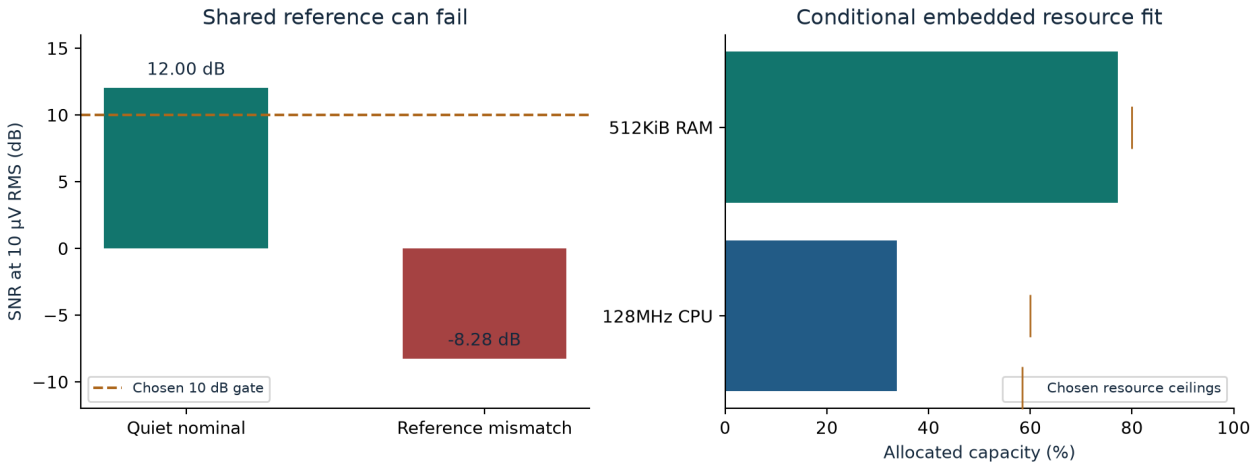
Selected access: intradural surface recording plus separate epidural stimulation. Sixteen recorded streams, two current sources and independent delivered-output monitoring. Application modes are initially exclusive; access and recruitment remain target-specific.

The companion has a separately allocated 5 V/1 A external source and 1 W base budget. The external pressure instrument and offline Studio are separately powered. Portable consumer sensing and transcutaneous charging losses remain open.

What has been established here: reproducible arithmetic, explicit data contracts, conditional nominal fit and rejected resource combinations. **What has not:** human neural observability, retained-learning benefit, safe electrode chemistry, complete circuits, populated implant CAD or chronic qualification.

AstroSam / Mind-Link is credited for the future-concept-to-inspectable-design approach. The SCCI mechanism, source audit, limits and engineering work are Project ATO contributions.

01 / Recording and computation

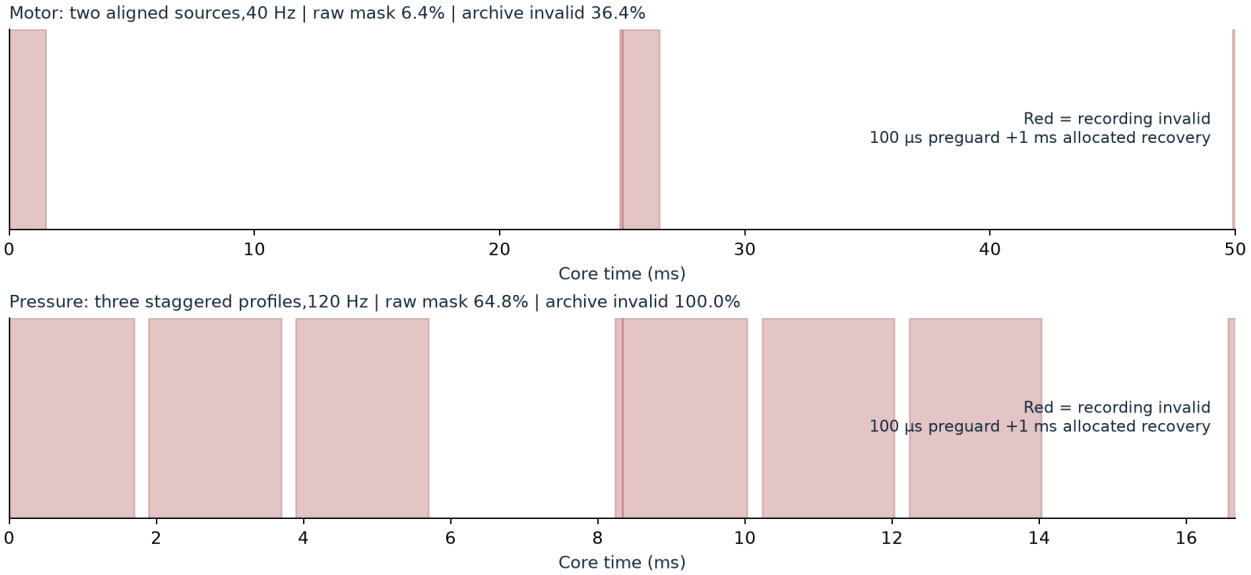


Resource	Selected budget / result
RHD2216 witness	16 differential streams; one multiplexed 16-bit ADC; 4 ksample/s per stream
Neural SPI / apertures	8 MHz; 19 slots ×3 µs=57 µs per 250 µs scan; 45 µs aperture span
nRF5340 application core	128 MHz; 43.1986 M cycles/s estimated,33.75% including contingency
Application RAM	404,992 /524,288 bytes; 22.75% spare; 2 s archive ring 192,000 bytes
Application flash	827,392 /1,048,576 bytes; 21.09% spare; single firmware image
Normal aggregate telemetry	86,528 bit/s including 30% headroom; 160 kbit/s chosen service target

The quiet shared-reference case passes a chosen 10 dB screening gate. The high-impedance mismatch case fails it. Typical chip noise/CMRR are not guaranteed bounds. Neural validity can require local bipolar/reference redesign; an assumed decoder cannot repair missing information.

The same workload fails a 60% CPU budget at 64 MHz. A 2 s raw-rate ring needs 596,992 bytes and does not fit. Continuous raw 4 k recording is 1.536 Mbit/s and exceeds the 1 Mbit/s target. Network flash/RAM leave 31.25%/25% spare under separate allocations. SPIM1 reserves 128 µs wire +122 µs local work for output admission. Actual images, execution and RF remain unverified.

02 / Causal timing and artifacts

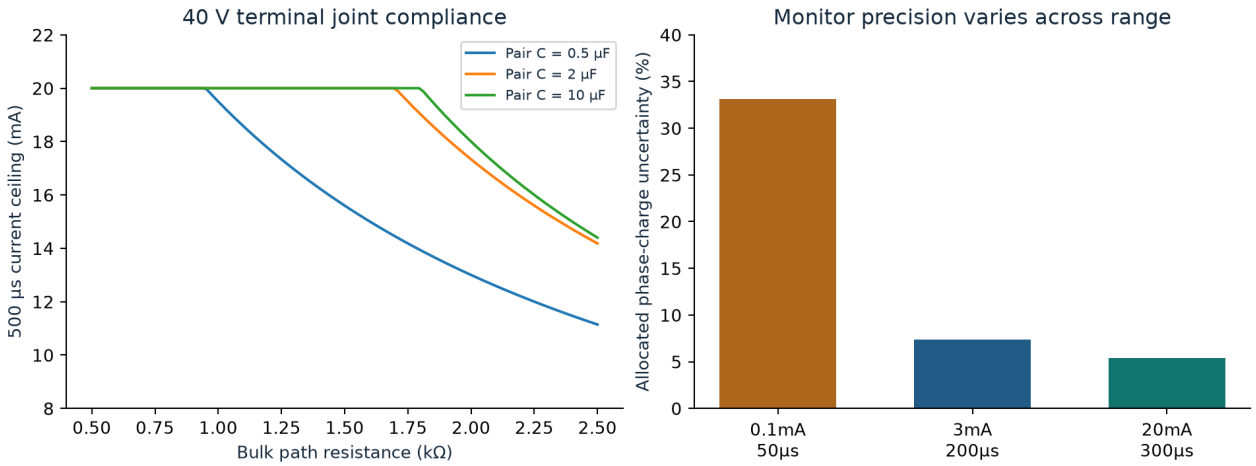


Timing quantity	Conditional allocation
Unframed fast chain	11.85 ms, with 200 Hz dispatch and 0.75 ms policy execution
20 ms uniform history midpoint at delivery	21.85 ms; actual masks/feature weighting change age
Archive FIR added to midpoint age	25.60 ms total; archive excluded from fast motor loop
Free-running 40 Hz frame	36.85 ms worst reactive bound; fails 25 ms chosen target

Known paced goals can be pre-admitted. Event anchoring must respect per-profile rate limits and physical-source availability. Pressure source 0 serves three 120 Hz profiles at 360 events/s: 0.95 ms service fits 2 ms minimum spacing. This allocation does not qualify electrode reset or guarantee a response to an unseen perturbation.

Pressure stimulation leaves only 35.2% average usable raw recording in this mask model. At 3 ms recovery the mask grows to 93.6%; at 5 ms it covers the frame. Use external pressure feedback in this mode. A 1 ms amplifier recovery allocation is not electrode reset or proven hardware recovery. The archive FIR uses 7.5 ms support: any invalid raw input invalidates its output. This masks 36.4% of motor archive and 100% of pressure archive; recursive-feature reset/warm-up also needs testing. Validity must qualify release.

03 / Output and delivered charge

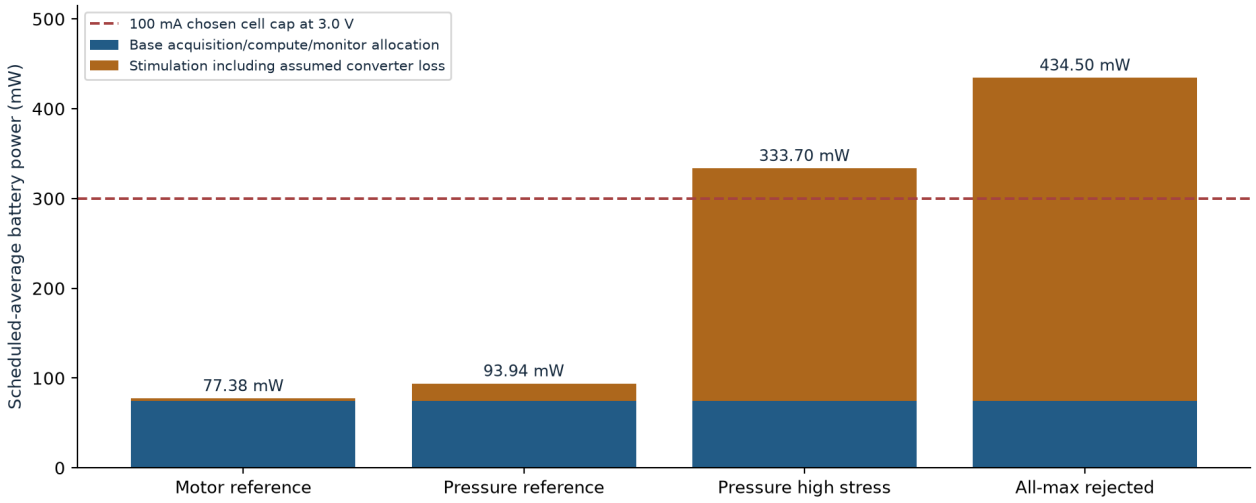


Output allocation	Meaning / limit
Two independent sources	16 routable contacts +2 dedicated returns; at most 2 concurrent
0.1–20 mA; 50–500 µs/phase	Joint ceiling 9 µC planned,10 µC including delivery uncertainty
48 V rail /40 V usable terminal	4 V pair-polarization reserve: 20 mA through 2 kΩ needs 44 V and fails
Independent ADS7953 monitor	450 ksample/s; 16 MHz separate SPI; 49.5% occupancy with chosen overhead
Effective HV reservoir 33 µF	Two 20 mA/500 µs symmetric pulses: 40 µC draw,1.212 V droop

The waveform is an ATO symmetric active biphasic choice with 100 µs interphase. Matching a study pulse-width number does not prove equivalent recruitment or recharge. Equal commands are insufficient: integrate actual phase current and include clipping/faults.

Charge density is a geometric quantity, not a safe threshold. At 6 mm²,6 µC gives 100 µC/cm² and 10 µC gives 166.7 µC/cm². Contact material, nonlinear polarization and chronic chemistry remain unqualified. Allocated monitor errors apply to a constant-phase witness, not unbounded inter-sample spikes.

04 / Power and stored energy

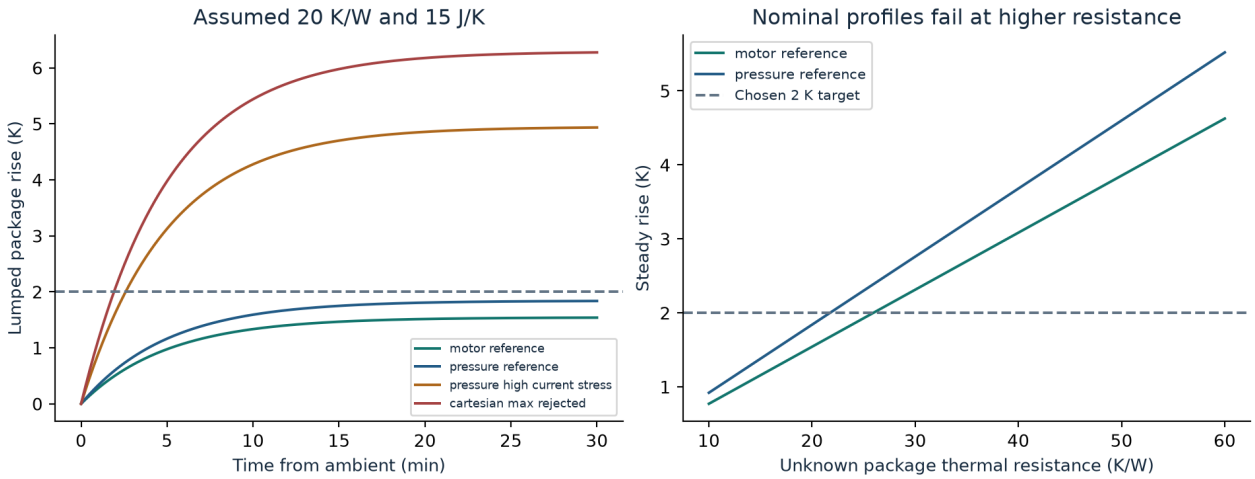


Reference schedule	Battery mW	Runtime h	Continuous gates
Motor reference	77.38	7.44	Conditional pass
Pressure reference	93.94	6.13	Conditional pass
Pressure high stress	333.70	1.73	Reject
All-max corner	434.50	1.33	Reject

Base 74.5 mW includes AFE 4.5, compute/network 35, monitor/analog 20, LV regulation 5, driver quiescent 8 and boost overhead 2 mW. Converter efficiency is an assumed 80%. All are engineering allocations with unresolved component/firmware behavior.

Contego 200 provides a 3.6 V nominal 200 mAh capacity witness atC/5 and 20 °C. Chosen usable energy is 80% ×0.72 Wh=0.576 Wh. Actual runtime depends on load, temperature and cell history. Its negative case requires isolation. High pressure stress draws 111.23 mA at 3.0 V and fails the selected 100 mA average-current cap.

05 / Heat, recovery and charging



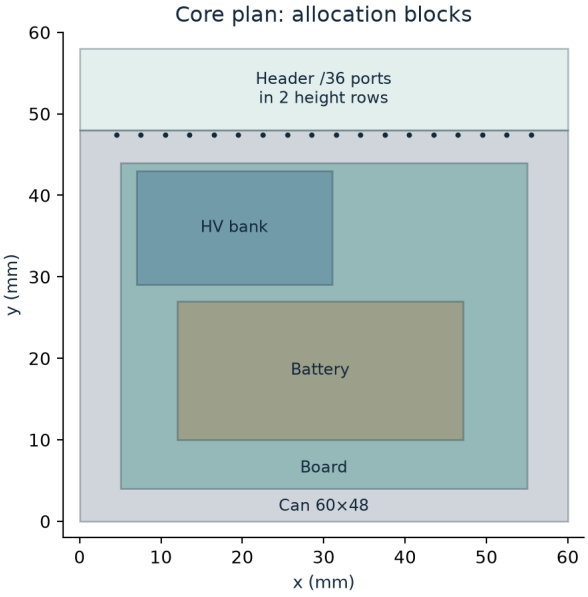
Package heat is conservatively battery input minus modeled resistive electrode-load deposition. It is not a tissue hotspot model. The chosen 2 K rise target is an engineering constraint, not an established clinical safety threshold.

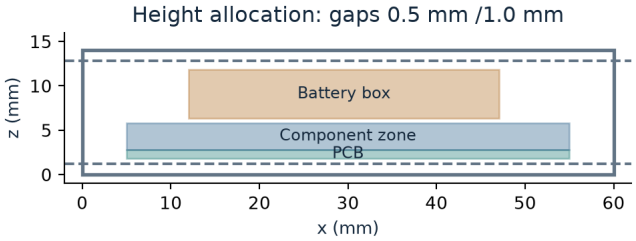
At an assumed 20 K/W, nominal motor/pressure rise is 1.54/1.84 K. Pressure requires ≤21.74 K/W for this target. Both references can fail at 40 or 60 K/W; nominal closure depends on unknown package coupling. The stronger pressure case reaches 2 K in about 155 s from ambient in this lumped model. A warm start shortens that interval.

Separate unresolved process	Calculated illustration
Electrode residual recovery	Two 5 µF interfaces,1 kΩ path: 4 V→20 mV in 13.25 ms, even with ideal shorting
Idle charging at 20 mA /4 V	85% assumed receiver-to-cell efficiency +10 mW idle: 24.12 mW case heat; 10 h capacity/current before taper

Thermal coupling, coil/RF losses, charging alignment, nonlinear chemistry, phase-trip response and power-loss balancing remain later circuit/field/biological checks. This stage does not silently assume energy recovery or successful passive discharge.

06 / A preliminary physical body





Allocation	Nominal envelope
Core can + separate header	60×48×14 mm metal-like can plus 60×10×14 mm dielectric allocation
Board / component zone	50×40×1 mm PCB with 3 mm component-height reservation
Battery / HV bank	35.1×17×5.5 mm bounding box; 24×14×3 mm effective-capacitor reservation
Interfaces / companion	36 illustrative feedthrough ports; 100×60×24 mm companion

The parametric STEP source contains labelled solids, an open can and separate unsealed lid, board/battery/HV reservations, header, port bores/insulators/pins and companion. Nominal major-solid intersections and BRep validity are checked.

These are allocation models. The actual populated KiCad assembly, lead/contact geometry, tolerance stack, cell insulation, swelling, welds, qualified feedthroughs, strain relief, anatomical fit and antenna/coil must still determine the final body. A metal can does not establish RF transmission or thermal resistance.

07 / What this stage establishes

Status	Concrete outcome
Calculated and reproducible	Signal/noise; sampling/data; analytical compute/RAM; delivery-monitor capacity; blanking/latency; compliance/charge; energy/heat and envelope fit
Specified	Eleven versioned control contracts; freshness/validity; output admission; two exclusive modes; assistance-off assessment
Rejected explicitly	64 MHz workload; 2 s raw 4 k ring; continuous raw 4 k target; all-max output; sustained high-pressure pattern in this package; neural fast mode during dense pressure output
Unestablished	Actual neural information, selective recruitment, retained learning, safe chemistry, chronic packaging, full circuit/firmware operation, RF and real thermal coupling

Next: computational mechanism. Generate electrode observations from a declared spinal population model; decode observable samples without hidden state; identify uncertain learner response; connect bounded commands to delivered current and modeled body response. Compare neural/external and expert/learner policies, with zero-benefit and harmful coupling cases. Electronics and populated engineering CAD follow these shared contracts.

Reproduction: scripts/run_architecture_budgets.py; inputs/results/source audits under docs/stage 2 and results/stage 2. Full parameter provenance and unresolved gates are in the companion Markdown/JSON dossier.

Selected primary / manufacturer sources

Intan RHD2000 architecture/specifications

Nordic nRF5340 product specification

TI ADS7953 delivery converter

Analog Devices LT8365 power witness

Resonetics Contego 200 battery datasheet

Phillips et al.2025: systolic-pressure feedback

Leung et al.2015: in-vivo charge injection